12. Which of the following statements is incorrect?
    (a) The higher the vapour pressure, the lower is the boiling point.
    (b) The higher the boiling point, the greater is the intermolecular force of attraction.
    (c) Vapour pressure and boiling point are not related to each other.
    (d) Boiling point of a liquid can be reduced by decreasing external pressure.

13. Which of the following is a homogeneous mixture?
    (a) Muddy water
    (b) Air
    (c) Finely powdered chalk in water
    (d) Mixture of saw dust and iron powder

14. The presence of impurities _____ of a liquid.
    (a) increases the boiling point
    (b) decreases the boiling point
    (c) cannot alter the boiling point
    (d) increases the freezing point

15. The melting point of a solid can be increased by increasing pressure on it. Which of the following conclusions can be drawn from this?
    (a) The solid expands on melting.
    (b) The solid contracts on melting.
    (c) The density of the solid is very high.
    (d) The solid contains some impurities.

Space for rough work

Answer Keys

Assessment Test I

1. (c) **2.** (c) **3.** (a) **4.** (d) **5.** (d) **6.** (b) **7.** (d) **8.** (b) **9.** (b) **10.** (b)
11. (a) **12.** (a) **13.** (b) **14.** (d) **15.** (c)

Assessment Test II

1. (c) **2.** (b) **3.** (c) **4.** (a) **5.** (c) **6.** (a) **7.** (c) **8.** (c) **9.** (b) **10.** (d)
11. (c) **12.** (c) **13.** (c) **14.** (a) **15.** (a)

Assessment Test III

1. (c) **2.** (c) **3.** (a) **4.** (a) **5.** (b) **6.** (d) **7.** (a) **8.** (c) **9.** (a) **10.** (a)
11. (a) **12.** (b) **13.** (a) **14.** (c) **15.** (b)

Assessment Test IV

1. (a) **2.** (d) **3.** (b) **4.** (c) **5.** (c) **6.** (c) **7.** (d) **8.** (b) **9.** (c) **10.** (b)
11. (c) **12.** (c) **13.** (b) **14.** (a) **15.** (a)

Language of Chemistry and Transformation of Substances 3

Reference: Coursebook - IIT Foundation Chemistry Class 8; Chapter - Language of Chemistry and Transformation of Substances; Page number - 3.1–3.24

Assessment Test I

Time: 30 min.

Space for rough work

Directions for questions from 1 to 15: Select the correct answer from the given options.

1. A metallic chloride of a trivalent metal M reacts with NH_4OH. Identify the balanced chemical equation.
   (a) $M_2Cl_3 + NH_4OH \rightarrow 3NH_4Cl + MOH$
   (b) $MCl_3 + NH_4OH \rightarrow MOH + NH_4Cl$
   (c) $M_3Cl_2 + NH_4OH \rightarrow 2NH_4Cl + M(OH)_3$
   (d) $MCl_3 + 3NH_4OH \rightarrow M(OH)_3 + 3NH_4Cl$

2. The formula of hydride of non-metal 'X' is XH_3. X can form different oxides in which X exhibits valencies one unit lower and one unit higher than that in XH_3. The formulae of the oxides are _____ and _____, respectively.
   (a) XO, XO_2 (b) X_2O, X_2O_3
   (c) XO_2, X_2O_3 (d) XO, X_2O_3

3. A trivalent metal forms salts with 'ic' and 'per' acids of chlorine. The number of oxygen atoms in the salts is respectively _____.
   (a) 9, 12 (b) 6, 8 (c) 8, 6 (d) 12, 9

4. A non-metal 'X' with electronic configurations 2, 4 forms two oxides in which 'X' shows valencies two and four. Oxide with higher valency is dissolved in water. The acid so produced is treated with NaOH to form salt. What could be the formula of the salt?
   (a) $NaXO_3$ (b) Na_2XO_3
   (c) $NaXO_4$ (d) Na_2XO_4

5. Both the metals 'X' and 'Y' can form two oxides in which their valencies differ by two units. What could be X and Y?
   (a) Cu, Fe (b) Fe, Pb (c) Pb, Sn (d) Sn, Cu

Space for rough work

6. A, B, C and D are four elements which have 1, 2, 3, and 6 electrons in valence M shell. The four elements combine with oxygen to form compounds P, Q, R, and S, respectively. Arrange P, Q, R and S in the decreasing order of number of oxygen atoms which combine with one atom of the corresponding element.
(a) SRQP (b) SRPQ (c) RSQP (d) RSPQ

7. $NaOH + HClO_4 \rightarrow$ salt 1
$Ca(OH)_2 + 2HClO_3 \rightarrow$ salt 2
$Pb(OH)_4 + 4HClO_2 \rightarrow$ salt 3
$Fe(OH)_3 + HOCl \rightarrow$ salt 4
Arrange the salts in the correct order of their number of oxygen atoms.
(a) Salt 3 > salt 2 > salt 1 > salt 4
(b) Salt 3 > salt 2 > salt 4 > salt 1
(c) Salt 2 > salt 3 > salt 1 > salt 4
(d) Salt 2 > salt 3 > salt 4 > salt 1

8. Limestone on heating gives solid 'X' which on dissolution in water forms 'Y'. Formation of X and Y are ______ and ______ reactions, respectively.
(a) endothermic, compound–compound combination
(b) exothermic, compound–compound combination
(c) combination, exothermic
(d) endothermic, displacement

9. Metal M forms an oxide with formula M_2O. What is the formula of the compound formed between M and a non-metal A with six electrons in its M shell?
(a) M_3A_2 (b) M_2A (c) MA_2 (d) M_2A_3

10. Metal with electronic configuration 2, 8, 14, 2 can form two different hydroxides A and B. Both the hydroxides on reaction with sulphuric acid give two salts 'X' and 'Y'. The two salts X and Y are ______ and ______, respectively.
(a) MSO_3, $M_2(SO_3)_3$ (b) MSO_4, $M_2(SO_4)_3$
(c) M_2SO_4, MSO_4 (d) M_2SO_4, $M_2(SO_4)_3$

11. Match the entries of Column A with those of Column B.

Column A (Compound)		Column B Valency of +ve radical	–ve radical
(i) Cupric phosphate	(A)	2	2
(ii) Stannic sulphite	(B)	4	3
(iii) Barium sulphate	(C)	2	3
(iv) Plumbic phosphate	(D)	4	2

Space for rough work

(a) (i) → (D); (ii) → (C); (iii) → (B); (iv) → (A)
(b) (i) → (B); (ii) → (C); (iii) → (D); (iv) → (A)
(c) (i) → (C); (ii) → (D); (iii) → (B); (iv) → (A)
(d) (i) → (C); (ii) → (D); (iii) → (A); (iv) → (B)

12. The molecular mass of oxide of a metal M is 40 and percentage of metal is 60%. What is the molecular mass of perchlorate of the metal?
(a) 191 (b) 223 (c) 159 (d) 255

13. The ratio by weight of oxygen that combines with fixed weight of nitrogen in NO, NO_2, and N_2O_3 is _____.
(a) 2:4:3 (b) 1:2:3 (c) 2:3:4 (d) 2:1:3

14. **Assertion (A):** The ratio of number of metal atoms to total number of atoms in ferric hydrogen phosphate is 1:10.
Reason (R): Ferric hydrogen phosphate contains trivalent positive and monovalent –ve ion.
(a) Both A and R are true and R is the correct explanation of A.
(b) Both A and R are true, but R is not the correct explanation of A.
(c) A is true and R is false.
(d) A is false and R is true.

15. **Assertion (A):** Green colour of $FeSO_4$ solution fades away when Zn rod is dipped in it but not Ni rod.
Reason (R): The order of reactivity is Ni > Zn > Fe.
(a) Both A and R are true and R is the correct explanation of A.
(b) Both A and R are true, but R is not the correct explanation of A.
(c) A is true and R is false.
(d) A is false and R is true.

Assessment Test II

Time: 30 min.

Space for rough work

Directions for questions from 1 to 15: Select the correct answer from the given options.

1. Bivalent metal hydroxide reacts with perchloric acid to form corresponding salt and water. Identify the balanced chemical equation.
   (a) $M(OH)_2 + 2HClO_4 \rightarrow M(ClO_4)_2 + 2H_2O$
   (b) $M(OH)_2 + 2HClO_3 \rightarrow M(ClO_3)_2 + 2H_2O$
   (c) $M(OH)_3 + 3HClO_4 \rightarrow M(ClO_4)_3 + 3H_2O$
   (d) $M(OH)_3 + 3HClO_3 \rightarrow M(ClO_3)_3 + 3H_2O$

2. The formula of hydride of non-metal Y is H_2Y. Y can form different oxides in which Y shows valencies which differ by two units. The difference in valencies of Y in its lower oxide and hydride is also 2. Then, the formulae of the two oxides are _____ and ______.
   (a) YO, YO_2 (b) YO_2, YO_3
   (c) Y_2O_3, Y_2O_5 (d) YO_2, Y_2O_3

3. The monovalent positive radical has four hydrogen atoms and a bivalent negative radical has four oxygen atoms. What could be the formula of the salt formed by these two radicals?
   (a) $AH_4(BO_3)_2$ (b) $(AH_4)_3BO_4$
   (c) AH_3BO_4 (d) $(AH_4)_2BO_4$

4. An acid contains three hydrogen atoms, one non-metal 'X', and four oxygen atoms. What could be the formula of the salt formed when the acid is treated with bivalent metal hydroxide?
   (a) $M_2(XO_4)_3$ (b) MXO_4
   (c) $M_3(XO_4)_2$ (d) $M(XO_4)_2$

5. Both metals 'A' and 'B' can form two oxides, and their valencies differ by one unit. What could be A and B?
   (a) Cu, Hg (b) Sn, Cu (c) Pb, Hg (d) Pb, Sn

6. Non-metal A forms four oxides P, Q, R and S in which the valencies of A are 3, 2, 5 and 1, respectively. Arrange P, Q, R and S in the descending order of number of oxygen atoms which combine with one atom of non-metal A.
   (a) R, P, Q, S (b) P, R, Q, S
   (c) R, P, S, Q (d) P, R, S, Q

Space for rough work

7. Arrange the carbonate, sulphate, chlorite, and hydroxide radicals in the correct order of number of oxygen atoms.
   (a) Chlorite > carbonate > sulphate > hydroxide
   (b) Carbonate > hydroxide > chlorite > sulphate
   (c) Hydroxide < chlorite < carbonate < sulphate
   (d) Sulphate > chlorite > carbonate > hydroxide

8. Slaked lime turns milky on treatment with carbon dioxide. The reaction is a _____ and _____ reaction.
   (a) Precipitation, neutralisation
   (b) Precipitation, redox
   (c) Double displacement, redox
   (d) Redox, displacement

9. The formula of hydride of a metal 'X' is XH_2. What will be the formula of compound formed between X and a non-metal 'Y' with five electrons in its M shell?
   (a) X_2Y_3 (b) X_3Y_2 (c) XY_2 (d) X_2Y

10. The electronic configuration of a metal 'M' is 2, 8, 18, 1. What could be the formulae of nitrate and nitrides of metal M?
   (a) MNO_3, $M(NO_3)_2$, M_3N, M_3N_2
   (b) MNO_3, M_3N, $M_2(NO_3)_2$, M_3N_2
   (c) $M(NO_3)_2$, M_3N_2 only
   (d) MNO_3, M_3N_2 only

11. Match the entries of Column A with those of Column B.

Column A (Compound)	Column B (Total No. of Atoms)
(i) Mercuric oxide	(A) 8
(ii) Sodium peroxide	(B) 6
(iii) Ferrous sulphate	(C) 4
(iv) Sodium phosphate	(D) 2

   (a) (i) → (B); (ii) → (C); (iii) → (D); (iv) → (A)
   (b) (i) → (C); (ii) → (D); (iii) → (A); (iv) → (B)
   (c) (i) → (D); (ii) → (A); (iii) → (B); (iv) → (C)
   (d) (i) → (D); (ii) → (C); (iii) → (B); (iv) → (A)

12. Identify the compound with molecular weight 100.
   (a) H_2SO_4 (b) H_3PO_4
   (c) $CaCO_3$ (d) $Fe(OH)_3$

Space for rough work

13. The ratio by weight of oxygen that combines with fixed weight of nitrogen in three oxides is 1:5:2. The oxides are _________.
(a) N_2O_3, N_2O_5 and NO_2
(b) NO, N_2O_5 and NO_2
(c) N_2O, N_2O_3 and NO_2
(d) N_2O, N_2O_5 and NO

14. **Assertion (A):** In a compound, the total positive charge is equal to that of negative charge.
Reason (R): In a compound, the total number of atoms in positive ion is always equal to that of negative ion.
(a) Both A and R are true and R is the correct explanation of A.
(b) Both A and R are true, but R is not the correct explanation of A.
(c) A is true and R is false.
(d) A is false and R is true.

15. **Assertion (A):** Both Al and Zn liberate hydrogen with dilute hydrochloric acid.
Reason (R): Both Al and Zn are more reactive than hydrogen, and the reactivity order is Zn > Al > H.
(a) Both A and R are true and R is the correct explanation of A.
(b) Both A and R are true, but R is not the correct explanation of A.
(c) A is true and R is false.
(d) A is false and R is true.

Assessment Test III

Time: 30 min.

Space for rough work

Directions for questions from 1 to 15: Select the correct answer from the given options.

1. **Assertion (A):** Burning of a candle involves both physical and chemical changes.

   Reason (R): During the burning of a candle, the physical state and the composition change.

   (a) Both A and R are true and R is the correct explanation of A.
   (b) Both A and R are true, but R is not the correct explanation of A.
   (c) A is true and R is false.
   (d) A is false and R is true.

2. **Assertion (A):** The formula of calcium phosphide is Ca_2P_3.

   Reason (R): In a molecule of a compound, the total positive charges and the total negative charges are equal.

   (a) Both A and R are true and R is the correct explanation of A.
   (b) Both A and R are true, but R is not the correct explanation of A.
   (c) A is true and R is false.
   (d) A is false and R is true.

3. Arrange the following compounds in the increasing order of valency of positive radical.

   (A) Zn_3P_2 (B) Fe_2O_3
   (C) $PbCl_4$ (D) $NaNO_3$

   (a) ADCB (b) DABC
   (c) DACB (d) ADBC

4. Arrange the following reactions in the order of neutralization reaction, precipitation reaction, thermal decomposition and photolysis.

   (A) $NaCl + AgNO_3 \rightarrow AgCl \downarrow + NaNO_3$
   (B) $CaCO_3 \rightarrow CaO + CO_2$
   (C) $NaOH + HCl \rightarrow NaCl + H_2O$
   (D) $2HOCl \rightarrow 2HCl + O_2$

   (a) ACDB (b) CABD
   (c) CADB (d) ACBD

Space for rough work

5. Match the entries of Column A with those of Column B.

Column A	Column B
(i) Sodium chlorate	(A) $NaNO_3$
(ii) Sodium chlorite	(B) $NaNO_2$
(iii) Sodium nitrate	(C) $NaClO_4$
(iv) Sodium nitrite	(D) $NaClO_3$
	(E) $Na(NO_3)_2$
	(F) $NaClO_2$

(a) (i) → (D); (ii) → (F); (iii) → (B); (iv) → (A)
(b) (i) → (D); (ii) → (F); (iii) → (A); (iv) → (B)
(c) (i) → (F); (ii) → (D); (iii) → (A); (iv) → (B)
(d) (i) → (F); (ii) → (D); (iii) → (B); (iv) → (A)

6. Which among the following is a balanced chemical equation?
(a) $3CuO + 2NH_3 \rightarrow Cu + N_2 + 3H_2O$
(b) $2NH_3 + O_2 \rightarrow 2NO + 3H_2O$
(c) $2Ca_3(PO_4)_2 + 2SiO_2 \rightarrow 6CaSiO_3 + P_4O_{10}$
(d) $S + 2H_2SO_4 \rightarrow 3SO_2 + 2H_2O$

7. The atomic weight of a trivalent metal is 27. Calculate the molecular mass of its sulphate.
(a) 294 (b) 342 (c) 318 (d) 366

8. $H_2 + Cl_2 \rightarrow 2HCl$. The reaction is _________ as well as __________.
(a) synthesis, displacement (b) analysis, combination
(c) synthesis, combination (d) analysis, displacement

9. Study the reaction given below and calculate the weight of oxygen required to produce 45 g of NO?
$N_2 + O_2 \rightarrow NO$.
(a) 32 g (b) 24 g (c) 48 g (d) 56 g

10. In which among the following compounds, the metal shows maximum valency?
(a) Na_2O (b) ZnO (c) Fe_2O_3 (d) PbO_2

11. The ratio of weight of oxygen combining with fixed weight of sulphur in SO_2 and SO_3 is _____________.
(a) 3:2 (b) 2:3 (c) 2:1 (d) 1:2

12. Which among the following elements possesses maximum atomic mass?
(a) Al (b) S (c) Ca (d) O